MD ASIF BIN SAYED

+88 01722951139
asifbinsayed.duet@gmail.com
asifbinsayed
Md Asif Bin Sayed

Objective

I am a final-year undergraduate in Electrical and Electronic Engineering with a strong foundation in nanotechnology, nanoelectronics, and their applications in advanced device architectures and biomedical systems. My research background spans image-vision–based ABEL trapping, autonomous robotics, and machine learning, which has equipped me with analytical, computational, and experimental skills. I am highly motivated to expand my expertise into nanomaterials, flexible electronics, and nanoscale device design, and aspire to contribute to pioneering research in these areas at the postgraduate level.

Education

Bachelor of Science | ***Electrical and Electronic Engineering*** — Dec. 2022 – Present
Dhaka University of Engineering and Technology (DUET) — Gazipur, Bangladesh

Diploma in Engineering | ***Electronics Technology*** — Aug. 2016 – Aug. 2020
Gopalganj Polytechnic Institute — Chandradigholia, Gopalganj, Bangladesh
Grade: 3.61 Out of 4.00

Secondary School Certificate | ***Science Group*** — Passing Year: 2016
Bheramara High School — Bheramara, Kushtia, Bangladesh
GPA: 4.56 Out of 5.00

Work Experience

Senior Vice President — December 2025 – Present
DUET Robotics Club (DRC) — Gazipur, Bangladesh

Campus Ambassador — March 2025 – March 2026
Ulkasemi Pvt. Limited

Finance Secretary — January 2025 – December 2025
DUET Robotics Club (DRC) — Gazipur, Bangladesh

Instructor — Present
Softmax Online School

Assistant General Secretary — 2024 – 2025
DUET Robotics Club (DRC) — Gazipur, Bangladesh

Instructor, Robotics — 2023 – Present
DUET Robotics Club (DRC) — Gazipur, Bangladesh
Conducted hands-on training on robotics, embedded systems, and autonomous navigation.

Tutor, Engineering University Admission Preparation — 2022 – 2023
Private Coaching Center — Gazipur, Bangladesh

Skills

Languages: English (Good reading; moderate listening and speaking), Bangla (Native proficiency)
Programming Languages: Python, C, MATLAB, Arduino
Data Science & Machine Learning: TensorFlow, PyTorch, Keras, Scikit-Learn, Pandas, NumPy
Cloud Technologies: Google Cloud Platform
Specialized Areas: Robotics, Image Processing, Sensor Works, Machine Learning
Mathematical & Statistical Tools: COMSOL Multiphysics, MATLAB, MS Word, Excel
Other Tools & Technologies: OrCAD, PSpice, Raspberry Pi, YOLO
Research Skills: Data Forecasting, Signal Processing, Robotics Prototyping, Project Simulation, Model Training, Literature Review
Web Technologies: Blogspot

Research Work

Early Stage Cervical Cancer Detection Using Deep Learning Ongoing
Machine Learning, Deep Learning, Medical Image Analysis

Portable Non-Invasive Cardiac Pacing System Ongoing
Embedded Systems, Biomedical Engineering, Control Systems

Solar Storm Prediction using LSTM-CNN Hybrid Model Completed
Machine Learning, Deep Learning, Time-Series Forecasting

Developing a hybrid model combining LSTM and CNN for solar storm prediction using solar activity data and geomagnetic parameters.

Focus on time-series forecasting and feature extraction using deep learning techniques (TensorFlow, Keras).

Nano Particle Trapping Methodology Completed
Image Processing, Nanomaterials

Analyzing nanoparticle trapping using image processing based on HSV color space to optimize trapping efficiency and conditions.

Leveraging data-driven insights with TensorFlow and PyTorch to refine trapping parameters and performance.

Autonomous Trash Collection Robot Completed
Machine Learning, Computer Vision, Autonomous Systems

Developed a robot integrating ML algorithms (TensorFlow) for trash classification, object recognition, and obstacle avoidance.

Focus on combining lane detection, YOLOv11 for trash classification, and GPS-based navigation.

Research Interests

Nanotechnology and Nanoelectronics
Biomedical Applications of Nanoelectronics
Machine Learning and Deep Learning
Data-Driven Research and Intelligent Autonomous Systems
Application of ML/DL in Real-World Problems using *TensorFlow, Keras, PyTorch, and Scikit-Learn*
Robotics Prototyping
Early-Stage Disease Detection using Machine Learning Techniques

Projects

Plant Nutrient Deficiency Detection System | *Python, OpenCV, Arduino, GSM* 2025

Remote Weather Telemetry Using LoRa Communication | *LoRa,Custom Data Encryption* 2024

Autonomous Trash Collector Robot | *Python, YOLO, Raspberry Pi* 2024

Mars Rover Navigation System | *Python, OpenCV, Machine Learning* 2024

Autonomous Firefighting Robot | *Arduino, Flame Sensors, Motor Controllers* 2024

Advanced Line Following Robot | *Arduino, IR Sensors, Sonar Sensors* 2023

Maze Solving Robot | *Python, Arduino, Sensors* 2023

IR Sensor People Counter | *Arduino, Sensors* 2022

Honors and Awards

Champion, Poster Presentation January 2025
Industrial Expo and Job Fair 2025, DUET

2nd Place, Cozmoclench October 2024
Techfest IIT Bombay Bangladesh Zonal Round

4th Place, Meshmerize October 2024
Techfest IIT Bombay Bangladesh Zonal Round

Champion, Advanced Line Following Robot July 2023
DUET Techfest 2023

Participant, Technoxian Bangladesh National Round March 2024
Technoxian World Robotics Championship

Volunteer Experience

Organizer 2025
DUET Techfest 2025 Gazipur, Bangladesh

Meshmerize Segment Organizer October 2024
Techfest, IIT Bombay Bangladesh Zonal 2024

Volunteer April 2024
3rd International Conference on Advancement in Electrical and Electronic Engineering, ICAEEE 2024

LFR Segment Organizer 2024
DRC Fest 2024

References

1. **Dr. Md. Mahmudur Rahman**
   Professor, Department of Electrical and Electronic Engineering
   Dhaka University of Engineering & Technology, Gazipur
   Email: m.rahman@duet.ac.bd
   Phone: +88 01308496660
   Relationship: Research Supervisor

2. **Mr. Sanjid-E-Elahi**
Assistant Professor, Department of Electrical and Electronic Engineering
Dhaka University of Engineering & Technology, Gazipur
Email: sanjid@duet.ac.bd
Phone: +88 01775911493
Relationship: Course Teacher